THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a : b = 4 : 9$ and $b : c = 6 : 5$ and $a + b + c = 369$, then b equals: 1

(a) 162 **(b)** 171 **(c)** 135 **(d)** 72

2. 90 men have provisions for 24 days. After 6 days 30 men leave. The remaining provisions last the rest of the men (days): 1

(a) 29 **(b)** 36 **(c)** 18 **(d)** 27

3. A vessel has 45 L milk and 20 L water. The litres of water to be added to make the ratio of milk to water 3:2 is: 1

(a) 25 **(b)** 10 **(c)** 20 **(d)** 13

4. The wages of 8 workers for 25 days are ₹ 36000. The wages of 15 workers for 12 days at the same daily rate (in ₹) are: 1

(a) 67500 **(b)** 32400 **(c)** 36000 **(d)** 32580

5. The present ages of Ria and Sia are in the ratio 4:7. After 10 years the ratio will be 2:3. Ria's present age (years) is: 1

(a) 30 **(b)** 18 **(c)** 35 **(d)** 20

6. ₹ 5400 is divided among three friends in the ratio 2:7:9. The difference between the largest and the smallest share (in ₹) is: 1

(a) 600 **(b)** 2101 **(c)** 2100 **(d)** 2700

7. In a pie chart a 54° sector represents 21 students. The total number of students is: 1

(a) 70 **(b)** 756 **(c)** 161 **(d)** 140

8. If $a : b = 5 : 7$ and $b : c = 14 : 9$ and $a + b + c = 264$, then b equals: 1

(a) 120 **(b)** 72 **(c)** 112 **(d)** 80

9. **Assertion (A):** On a map with the scale 1 cm : 5 km, a distance of 4 cm represents 20 km. 1
Reason (R): The scale 1 cm : 5 km means that 5 cm on the map represents 1 km on the ground.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** If x and y are inversely proportional, then $\frac{x}{y}$ is constant. 1

Reason (R): If x and y are inversely proportional, then xy is constant.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two numbers are in the ratio $5 : 2$ and their difference is 45. Find the numbers. 2

12. The present ages of a father and son are in the ratio 8:3. After 10 years the ratio will be 2:1. Find their present ages. 2

13. If $x : y = 7 : 3$, find the ratio $(2x + 5y) : (3x - 4y)$. 2

14. Two numbers are in the ratio $9 : 4$ and their difference is 60. Find the numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A garrison of 300 men has provisions for 30 days. After 6 days 100 men leave. For how many more days will the remaining provisions last for the men who stay? 3

16. A mixture contains milk and water in the ratio 4:5. When 8 L of water is added, the ratio of milk to water becomes 4:7. Find the quantity of milk and water in the original mixture. 3

17. In a pie chart, the sector for Science is 24° and represents 8 students, while the sector for Maths is 60°. Find the total number of students and the number who like Maths. 3

OR

A garrison of 160 men has provisions for 50 days. After 20 days 40 men leave. For how many more days will the remaining provisions last for the men who stay? 3

18. A mixture contains milk and water in the ratio 7:3. When 15 L of water is added, the ratio of milk to water becomes 7:8. Find the quantity of milk and water in the original mixture. 3

OR

In a pie chart, the sector for Science is 36° and represents 13 students, while the sector for Maths is 108°. Find the total number of students and the number who like Maths. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** 15 workers can build a village road in 24 days, working at the same rate. 5
(i) Find the total work in worker-days.
(ii) In how many days will 20 workers complete the road?
(iii) In how many days will 9 workers complete it?
(iv) How many more workers are needed to finish the road in 12 days?

20. The present ages of Anil and Bina are in the ratio 2:5. After 6 years the ratio of their ages will be 5:11. Find their present ages and the ratio of their ages 6 years ago. 5

OR

36 workers can finish a job in 20 days. After 5 days 9 workers leave. In how many total days will the job be finished? By how many days is it delayed? 5

END OF QUESTION PAPER — Total: 40 Marks